Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Akandwanaho, David
Date of birth:	2/2/1987
Subject:	040-Human Performance
Result:	37,5 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	163	40.	HUMAN PERFORMANCE	1,00	1,00
2	179	40.	HUMAN PERFORMANCE	0,00	1,00
3	162	40.	HUMAN PERFORMANCE	0,00	1,00
4	175	40.	HUMAN PERFORMANCE	0,00	1,00
5	174	40.	HUMAN PERFORMANCE	1,00	1,00
6	206	40.	HUMAN PERFORMANCE	0,00	1,00
7	194	40.	HUMAN PERFORMANCE	0,00	1,00
8	165	40.	HUMAN PERFORMANCE	1,00	1,00
9	195	40.	HUMAN PERFORMANCE	1,00	1,00
10	168	40.	HUMAN PERFORMANCE	1,00	1,00
11	209	40.	HUMAN PERFORMANCE	1,00	1,00
12	198	40.	HUMAN PERFORMANCE	0,00	1,00
13	103	40.2	Basic aviation physiology and health maintenance	0,00	1,00
14	68	40.2.1.1	The atmosphere	0,00	1,00
15	34	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
16	24	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
17	47	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
18	99	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
19	12	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
20	63	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
21	42	40.2.2.2	Vision	0,00	1,00
22	13	40.2.2.2	Vision	0,00	1,00
23	36	40.2.2.2	Vision	1,00	1,00
24	27	40.2.2.2	Vision	1,00	1,00
25	96	40.2.2.3	Hearing	1,00	1,00
26	23	40.2.2.5	Integration of sensory inputs	0,00	1,00
27	58	40.2.2.5	Integration of sensory inputs	1,00	1,00
28	107	40.2.3	Health and hygiene	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	32	40.2.3.3	Problem areas for pilots	0,00	1,00
30	87	40.2.3.4	Intoxication	1,00	1,00
31	14	40.3	BASIC AVIATION PSYCHOLOGY	0,00	1,00
32	21	40.3.2.4	Error generation	0,00	1,00
33	111	40.3.3	Decision making	0,00	1,00
34	5	40.3.4.3	Co-operation	1,00	1,00
35	59	40.3.4.4	Communication	0,00	1,00
36	119	40.3.5	Human behaviour	0,00	1,00
37	114	40.3.5	Human behaviour	1,00	1,00
38	120	40.3.5	Human behaviour	0,00	1,00
39	1	40.3.5.1	Personality, attitude and behaviour	0,00	1,00
40	7	40.3.6.1	Arousal	0,00	1,00
			Total:38%	**15,00**	**40,00**

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Akandwanaho, David

Location: Uganda Civil Aviation Authority

Your Result: 15,00 from 40,00 Points (37,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	15,00	40,00	37,50
40.2. Basic aviation physiology and health maintenance	18	7,00	18,00	38,89
40.3. BASIC AVIATION PSYCHOLOGY	10	2,00	10,00	20,00
Total	**40**	**15,00**	**40,00**	**37,50**

1

Which among these is a result of hyperventilation?

- [x] **a lack of carbon dioxide in the body.**
- [ ] **breathing too rapidly causing a lack of oxygen.**
- [ ] **a need to increase the flow of supplemental oxygen.**

ID:	163(LTS:224854, LTM:163)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

2

Hazardous attitudes which contribute to poor pilot judgment can be effectively counteracted by

☒ **taking meaningful steps to be more assertive with attitudes.**

☐ **early recognition of these hazardous attitudes.**

☐ **an appropriate antidote.**

ID: 179(LTS:224868, LTM:179)
Syllabus reference: 40. HUMAN PERFORMANCE

Max. Points 1,00
Points 0,00

3

What physical change would most likely occur to occupants of an unpressurized aircraft flying above 15,000 feet without supplemental oxygen?

- ☒ **The pressure in the middle ear becomes less than the atmospheric pressure in the cabin.**
- ☐ **A blue coloration of the lips and fingernails develop along with tunnel vision.**
- ☐ **Gases trapped in the body contract and prevent nitrogen from escaping the bloodstream.**

ID:	162(LTS:224853, LTM:162)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	0,00

4

Examples of classic behavioral traps that experienced pilots may fall into are to

- ☐ assume additional responsibilities and assert PIC authority.
- ☐ complete a flight as planned, please passengers, meet schedules, and 'get the job done.'
- ☒ promote situational awareness and then necessary changes in behavior.

ID:	175(LTS:224865, LTM:175)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	0,00

5

One of the steps in the aeronautical decision making (ADM) process involved in good decision making is

- ☐ making a rational evaluation of the required actions.
- ☐ developing a 'can do' attitude.
- ☒ identifying personal attitudes hazardous to safe flight.

ID:	174(LTS:224864, LTM:174)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

6

What action should be taken if hyperventilation is suspected?

☒ Breathe at a slower rate by taking very deep breaths.

☐ Consciously force yourself to take deep breaths and breathe at a faster rate than normal.

☐ Consciously breathe at a slower rate than normal.

ID:	206(LTS:224891, LTM:206)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	0,00

7

What visual illusion creates the same effect as a narrower-than-usual runway?

- ☐ **An upsloping runway.**
- ☒ **A downsloping runway.**
- ☐ **A wider-than-usual runway.**

ID:	194(LTS:224879, LTM:194)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	0,00

8

A person should be able to overcome the symptoms of hyperventilation by

- ☐ **increasing the breathing rate**
- ☐ **refraining from the use of alcohol**
- ☒ **slowing the breathing rate**

ID:	165(LTS:224856, LTM:165)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

9

Abrupt head movement during a prolonged constant rate turn in IMC or simulated instrument conditions can cause

- ☐ **elevator illusion.**
- ☐ **false horizon.**
- ☒ **pilot disorientation.**

ID:	195(LTS:224880, LTM:195)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

10

What suggestion could you make to people who are experiencing motion sickness?

- ☐ Recommend taking medication to prevent motion sickness.
- ☒ Avoid unnecessary head movement and focus sight on a point outside the aircraft.
- ☐ Have the students lower their head, shut their eyes, and take deep breaths.

ID:	168(LTS:224859, LTM:168)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

11

What effect does haze have on the ability to see traffic or terrain features during flight?

- ☐ Haze causes the eyes to focus at infinity, making terrain features harder to see.
- ☒ Haze creates the illusion of being a greater distance than actual from the runway, and causes pilots to fly a lower approach.
- ☐ The eyes tend to overwork in haze and do not detect relative movement easily.

ID:	209(LTS:224894, LTM:209)	Max. Points	1,00
Syllabus reference:	40. HUMAN PERFORMANCE	Points	1,00

12

A rapid acceleration during takeoff can create the illusion of

☐ **being in a noseup attitude.**

☐ **diving into the ground.**

☒ **spinning in the opposite direction.**

ID: 198(LTS:224883, LTM:198)
Syllabus reference: 40. HUMAN PERFORMANCE

Max. Points 1,00
Points 0,00

13

Which of these is a common symptom of hyperventilation?

- ☐ **Drowsiness.**
- ☐ **A sense of well-being.**
- ☒ **Decreased breathing rate.**

ID:	103(LTS:224904, LTM:103)	Max. Points	1,00
Syllabus reference:	40.2 Basic aviation physiology and health maintenance	Points	0,00

14 **Fatigue and stress**

☒ **increase the tolerance to hypoxia**

☐ **will increase the tolerance to hypoxia when flying below 15 000 feet**

☐ **lower the tolerance to hypoxia**

☐ **do not affect hypoxia at all**

ID: 68(LTS:224976, LTM:68)
Syllabus reference: 40.2.1.1 The atmosphere

Max. Points 1,00
Points 0,00

15

Symptoms of decompression sickness

☒ **are only relevant when diving**

☐ **are flatulence and pain in the middle ear**

☐ **can only develop at altitudes of more than 40000 FT**

☐ **are bends, chokes, creeps and neurological symptoms**

ID:	34(LTS:224959, LTM:34)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	0,00

16

When faced with sustained cold temperature, how does the body resist this physical stress?

- ☐ **By vasodilatation which permits a greater flow of blood to the periphery.**
- ☐ **By increasing cardiac frequency.**
- ☐ **By intense vasoconstriction.**
- ☒ **By speeding up the metabolic rate in the Autonomic Nervous System.**

ID:	24(LTS:224956, LTM:24)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	0,00

17

The respiratory process consists mainly of

- ☐ **the transportation of oxygen to the cells and the elimination of carbon monoxide**
- ☐ **the transportation of carbon dioxide to the cells and elimination of oxygen**
- ☒ **the transportation of oxygen to the cells and the elimination of nitrogen**
- ☐ **the diffusion of oxygen through the respiratory membranes into the blood, transportation to the cells, diffusion into the cells and elimination of carbon dioxide from the body**

ID:	47(LTS:224969, LTM:47)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	0,00

18 **Hypoxia can be caused by**

☐ lack of nitrogen in ambient air

☐ increasing oxygen partial pressure used for the exchange of gases

☐ decreased ability of the haemoglobin to transport oxygen

☒ too much carbon dioxide in the blood

ID:	99(LTS:224951, LTM:99)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	0,00

19 **Hypoxia is caused by**

- ☐ an increased number of red blood cells
- ☐ reduced partial pressure of nitrogen in the lung
- ☒ reduced partial oxygen pressure in the lung
- ☐ a higher affinity of the red blood cells (haemoglobin) to oxygen

ID:	12(LTS:224957, LTM:12)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	1,00

20

When flying above 10.000 feet hypoxia arises because:

- ☐ composition of the air is different from sea level
- ☒ composition of the blood changes
- ☐ partial oxygen pressure is below the critical value of 55mm/Hg
- ☐ percentage of oxygen is lower than at sea level

ID:	63(LTS:224962, LTM:63)	Max. Points	1,00
Syllabus reference:	40.2.1.2 Respiratory and circulatory systems	Points	0,00

21

When flying through a thunderstorm with lightning you can protect yourself from flashblindness by:
1 - turning up the intensity of cockpit lights
2 - looking inside the cockpit
3 - wearing sunglasses
4 - using blinds or curtains when installed

- ☐ 1, 2 and 3 are correct, 4 is false
- ☐ 3 and 4 are correct, 1 and 2 are false
- ☐ 1, 2, 3 and 4 are correct
- ☒ 1 and 2 are correct, 3 and 4 are false

ID:	42(LTS:224911, LTM:42)	Max. Points	1,00
Syllabus reference:	40.2.2.2 Vision	Points	0,00

22

When focussing on near objects:

- ☐ **the pupil gets larger**
- ☐ **the shape of lens gets more spherical**
- ☐ **the cornea gets smaller**
- ☐ **the shape of lens gets flatter**

ID:	13(LTS:224915, LTM:13)	Max. Points	1,00
Syllabus reference:	40.2.2.2 Vision	Points	0,00

23

Presbyopia:

- ☐ is partial visual loss due to pressure changes in the eye.
- ☐ is caused by long-termed exposure to stimuli over 90dB.
- ☒ is common over the age of 50.
- ☐ surgical replacement of the lens the usual treatment and is compatible with flying.

ID:	36(LTS:224912, LTM:36)	Max. Points	1,00
Syllabus reference:	40.2.2.2 Vision	Points	1,00

24

Illuminated anti-collision lights in IMC

- ☐ can cause colour-illusions
- ☐ will effect the pilots binocular vision
- ☒ can cause disorientation
- ☐ will improve the pilots depth perception

ID:	27(LTS:224916, LTM:27)	Max. Points	1,00
Syllabus reference:	40.2.2.2 Vision	Points	1,00

25

Noise induced hearing loss:

☐ is not a permanent hearing loss, the nerve cells frequently recover.

☐ causes Eustationary Tube dysfunction.

☐ is also known as presbyacusis and is associated with pressure damage to the middle ear.

☒ is a condition resulting in permanent hearing loss of selected frequencies.

ID:	96(LTS:224994, LTM:96)	Max. Points	1,00
Syllabus reference:	40.2.2.3 Hearing	Points	1,00

26

For a pilot who is used to land on small and narrow runways only, approaching a larger and wider runway can lead to:

- ☐ steeper than normal approach dropping low
- ☒ risk to land short of the overrun
- ☐ flatter than normal approach with the risk of "ducking under"
- ☐ early or high "round out"

ID: 23(LTS:224924, LTM:23)
Syllabus reference: 40.2.2.5 Integration of sensory inputs

Max. Points 1,00
Points 0,00

27

What would be the effect if, in a tight turn, one bends down to pick up a pencil?

- ☐ **Inversion Illusion.**
- ☒ **Coriolis effect.**
- ☐ **Vertigo.**
- ☐ **Barotrauma.**

ID:	58(LTS:224925, LTM:58)	Max. Points	1,00
Syllabus reference:	40.2.2.5 Integration of sensory inputs	Points	1,00

28

Which of these is true regarding the presence of alcohol within the human body?

- ☐ An increase in altitude decreases the adverse effect of alcohol.
- ☐ A small amount of alcohol increases vision acuity.
- ☒ Judgment and decision-making abilities can be adversely affected by even small amounts of alcohol.

ID:	107(LTS:224902, LTM:107)	Max. Points	1,00
Syllabus reference:	40.2.3 Health and hygiene	Points	1,00

29

Flying with a "common cold":

- ☐ is permitted as long as you are on treatment with antibiotics.
- ☒ increases the risk of hypertension.
- ☐ will cause infection in other crew members if you are flying in a pressurised aircraft.
- ☐ may lead to incapacitation due to severe sinus or ear pain.

ID:	32(LTS:224899, LTM:32)	Max. Points	1,00
Syllabus reference:	40.2.3.3 Problem areas for pilots	Points	0,00

30

Which statement is correct?

1. Smokers have a greater chance of suffering from coronary heart disease
2. Smoking tobacco will raise the individuals physiological altitude during flight
3. Smokers have a greater chance of contracting lung cancer

- [ ] 1 and 3 are correct, 2 is false
- [ ] 1 and 2 are correct, 3 is false
- [x] 1,2 and 3 are correct
- [ ] 2 and 3 are correct, 1 is false

ID: 87(LTS:224977, LTM:87)
Syllabus reference: 40.2.3.4 Intoxication

Max. Points 1,00
Points 1,00

31

Carbon Monoxide is particularly dangerous because:

1. Its initial symptoms are not alarming
2. It is colourless
3. Its is odourless
4. It is highly toxic
5. Its effects are cumulative

- ☐ **all of the above**
- ☒ **2, 3, and 4 only**
- ☐ **2, 3, 4 and 5 only**
- ☐ **1, 2, 3 and 5 only**

ID:	14(LTS:224848, LTM:14)	Max. Points	1,00
Syllabus reference:	40.3 BASIC AVIATION PSYCHOLOGY	Points	0,00

32

Human error rates during the performance of a simple and repetitive task can normally be expected to be approximately:

- ☐ **1 in 500**
- ☐ **1 in 200**
- ☐ **1 in 100**
- ☒ **1 in 2000**

ID:	21(LTS:224982, LTM:21)	Max. Points	1,00
Syllabus reference:	40.3.2.4 Error generation	Points	0,00

33

Which of the following is condidered as a step involved in good Aeronautical Decision Making (ADM) process?

- ☐ **developing the 'right stuff' attitude.**
- ☒ **making a rational evaluation of the required actions.**
- ☐ **identifying personal attitudes hazardous to safe flight.**

ID: 111(LTS:225007, LTM:111)
Syllabus reference: 40.3.3 Decision making

Max. Points 1,00
Points 0,00

34

Which of the following statements best characterises a self-centered cockpit ?

- [x] **Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on**

- [ ] **The communication between crew members always increases when the captain takes charge of a situation**

- [ ] **While decreasing communication, the independence of each member bolsters the crew's synergy**

- [ ] **The egoistic and self-centered personality of the captain often leads to a synergetic cockpit**

ID:	5(LTS:224908, LTM:5)	Max. Points	1,00
Syllabus reference:	40.3.4.3 Co-operation	Points	1,00

35

"Stereotypes" are preconceptions or prejudices which can lead us to:

- ☐ **develop better teamwork by standardizing procedures.**
- ☒ **act in the same manner in all situations and thus assuring stability.**
- ☐ **mis-judge individuals even if we have contact with them.**
- ☐ **communicate non-verbally with a stranger.**

ID:	59(LTS:224971, LTM:59)	Max. Points	1,00
Syllabus reference:	40.3.4.4 Communication	Points	0,00

36

What is the first step in neutralizing a hazardous attitude in the ADM process?

☒ **Dealing with improper judgment.**

☐ **Recognition of invulnerability in the situation.**

☐ **Recognition of hazardous thoughts.**

ID:	119(LTS:224817, LTM:119)	Max. Points	1,00
Syllabus reference:	40.3.5 Human behaviour	Points	0,00

37

Some of these dangerous tendencies or behavior patterns which must be identified and eliminated include:

- [x] **Peer pressure, get-there-itis, loss of positional or situation awareness, and operating without adequate fuel reserves.**
- [ ] **Performance deficiencies from human factors such as, fatigue, illness or emotional problems.**
- [ ] **Deficiencies in instrument skills and knowledge of aircraft systems or limitations.**

ID:	114(LTS:224815, LTM:114)	Max. Points	1,00
Syllabus reference:	40.3.5 Human behaviour	Points	1,00

38

What should a pilot do when recognizing a thought as hazardous?

- ☒ **Develop this hazardous thought and follow through with modified action.**
- ☐ **Label that thought as hazardous, then correct that thought by stating the corresponding learned antidote.**
- ☐ **Avoid developing this hazardous thought.**

ID:	120(LTS:224818, LTM:120)	Max. Points	1,00
Syllabus reference:	40.3.5 Human behaviour	Points	0,00

39

Attitudes are defined as:

- ☐ **tendencies to respond to people, things or events in a particular manner**
- ☐ **the genetic predispositions for thinking and acting**
- ☒ **a synonym for behaviour**
- ☐ **the conditions necessary for carrying out an activity**

ID:	1(LTS:224808, LTM:1)	Max. Points	1,00
Syllabus reference:	40.3.5.1 Personality, attitude and behaviour	Points	0,00

40

An identical situation can be experienced by one pilot as exciting in a positive sense and by another pilot as threatening. In both cases:

- ☐ both pilots will experience the same amount of stress
- ☐ both pilots will loose their motor-coordination
- ☐ the arousal level of both pilots will be raised
- ☒ the pilot feeling threatened, will be much more relaxed, than the pilot looking forward to what may happen

ID:	7(LTS:224922, LTM:7)	Max. Points	1,00
Syllabus reference:	40.3.6.1 Arousal	Points	0,00